LAB No. 19

Backup recovery using cronjob in shell automation

Data loss due to accidental deletion, hardware failure, or system crashes is a critical concern in Linux systems. Regular backups and reliable recovery mechanisms are essential to ensure data safety and business continuity.

This lab focuses on **automating data backup and recovery in Linux using shell scripting and cron jobs**. Students will create a shell script to perform scheduled backups of important directories and store them in a secure backup location. The script will also support basic recovery of backed-up data. Automation using cron ensures backups occur at regular intervals without manual intervention.

LAB Objectives

- Understand the importance of data backup and recovery
- Create automated backup scripts using shell scripting
- Schedule backup tasks using cron jobs
- Perform data restoration from backups
- Maintain backup logs for verification and auditing

Tools and Software Required

Operating System

- Linux (Ubuntu / CentOS / Red Hat / Debian)

Backup and Recovery in Linux

Backup is the process of creating copies of data to prevent loss, while recovery is restoring data from backup copies. Linux supports various backup methods including full, incremental, and differential backups.

Shell Scripting for Backup Automation

Shell scripts allow administrators to:

- Automate repetitive backup tasks

- Include timestamped backup files
- Handle multiple directories
- Implement error checking and logging

Automation improves reliability and consistency.

Cron Job Scheduling

Cron is a time-based scheduler used to automate tasks such as backups.

Examples:

- Daily backups at midnight
- Weekly full backups

Cron ensures backups run even without user interaction.

Expected Outcomes

- Automated backup system using shell scripting
- Scheduled backups via cron job
- Successfully restored data from backup
- Log files recording backup activities

Student Task

Perform the following tasks briefly:

1. Create a shell script to back up a selected directory using tar or rsync.
2. Store backups in a designated backup location with date-based naming.
3. Add logging to record backup status and execution time.
4. Schedule the backup script using a cron job to run automatically (daily or weekly).
5. Simulate data loss and perform recovery using the backup files.
6. Verify restored data and document the results in the lab record.

OUTPUT RESULT:

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mkdir -p ~/important_data
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mkdir -p ~/backup_storage
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ echo "This is file 1" > ~/important_data/file1.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ echo "This is file 2" > ~/important_data/file2.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls ~/important_data
file1.txt  file2.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ nano backup.sh
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat backup.sh
#!/bin/bash

SOURCE="$HOME/important_data"
DEST="$HOME/backup_storage"

mkdir -p "$DEST"

DATE=$(date +"%Y-%m-%d_%H-%M-%S")
BACKUP_FILE="$DEST/backup_$DATE.tar.gz"
LOGFILE="$DEST/backup.log"

tar -czf "$BACKUP_FILE" "$SOURCE"

if [ $? -eq 0 ]; then
    echo "$(date): Backup SUCCESS -> $BACKUP_FILE" >> "$LOGFILE"
else
    echo "$(date): Backup FAILED" >> "$LOGFILE"
fi
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ chmod +x ~/backup.sh
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ./backup.sh
tar: Removing leading `/' from member names
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls ~/backup_storage
backup_2026-05-12_08-36-15.tar.gz  backup_2026-05-12_08-45-54.tar.gz  backup_2026-05-12_08-47-49.tar.gz  backup_2026-05-12_08-49-44.tar.gz  backup_2026-05-12_08-57-57.tar.gz  backup.log
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat ~/backup_storage/backup.log
Tue May 12 08:36:15 AM PKT 2026: Backup successful -> backup_2026-05-12_08-36-15.tar.gz
Tue May 12 08:45:54 AM PKT 2026: Backup failed
Tue May 12 08:47:49 AM PKT 2026: Backup successful -> backup_2026-05-12_08-47-49.tar.gz
Tue May 12 08:49:44 AM PKT 2026: Backup successful -> backup_2026-05-12_08-49-44.tar.gz
Tue May 12 08:57:57 AM PKT 2026: Backup SUCCESS -> /home/abdullah-sohail/backup_storage/backup_2026-05-12_08-57-57.tar.gz
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ crontab -e
No modification made
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ rm -r ~/important_data
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls ~/important_data
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ BACKUP_FILE=~/backup_storage/new_backup.tar.gz
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ tar -czf "$BACKUP_FILE" -C "$HOME" important_data
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls ~/backup_storage
backup_2026-05-12_08-36-15.tar.gz  backup_2026-05-12_08-47-49.tar.gz  backup_2026-05-12_08-57-57.tar.gz  new_backup.tar.gz
backup_2026-05-12_08-45-54.tar.gz  backup_2026-05-12_08-49-44.tar.gz  backup.log
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ tar -xzf ~/backup_storage/backup_2026-05-12_08-57-57.tar.gz
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls ~/home/abdullah-sohail/important_data
file1.txt  file2.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ BACKUP_FILE=~/backup_storage/final_backup.tar.gz
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ tar -czf "$BACKUP_FILE" -C "$HOME" important_data
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ rm -r ~/important_data
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ tar -xzf ~/backup_storage/final_backup.tar.gz -C ~/
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ tar -xzf ~/backup_storage/final_backup.tar.gz -C ~/
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls ~/important_data
file1.txt  file2.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat ~/important_data/file1.txt
This is file 1
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat ~/important_data/file2.txt
This is file 2
```